MILITARY SPECIFICATION

PERISCOPE XM33

1. SCOPE

1.1 This specification covers one type of infrared monocular periscope.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army, Frankford Arsenal

F10514542	Periscope XM33
F8565556	Pressure Tester, Two Station

FSC 1240

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PACKAGING DATA SHEET

10514542 Packaging of Periscope XM33

(Copies of specifications, standards, drawings and packaging data sheets required by the suppliers in connection with this specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The periscope shall be manufactured in accordance with Drawing F10514542 and drawings pertaining thereto.

3.2 General specifications. - The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926 and MIL-O-13830:

3.2.1 Specification MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.2.2 Specification MIL-O-13830.

(a) Cleanliness
(b) Illuminated reticle

3.3 Environmental.

3.3.1 Storage temperature. - The periscope shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all requirements of this specification at standard ambient temperature (60 to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of minus 65 degrees Fahrenheit and plus 155 degrees Fahrenheit,

3.3.2 Operating temperatures. - The periscope, shall meet the applicable requirements of 3.5.10 while exposed and thermally stabilized at ambient temperatures of minus 40 degrees Fahrenheit and plus 150 degrees Fahrenheit.

3.3.3 Vibration. - There shall be no visible damage or loosening of assembled parts as a result of vibration in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds with the input coupling immobilized and the periscope mounted in its normal operating position.

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3.4 Electrical check. - Electrical checks shall be met with the image converter tube, power supply and converter regulator or 1.5 volt "C" cell battery removed from the periscope. These checks may be conducted at any-time prior to final acceptance of periscope.

3.4.1 Dielectric strength. - There shall be no puncture of insulation, no visible evidence of damage due to corona,or decrease in dielectric strength due to the application of 20,000 ± 500 vdc to the circuit between pow supply and image converter tube for one minute.

3.4.2 Insulation resistance. - Insulation resistance between un-grounded circuits and the body or frame and between isolated circuits shall not be less than 2 megohms with 500 vdc applied.

3.4.3 Continuity. - The electrical circuit shall be checked for continuity by applying a maximum of 30 vdc and shall conform to the Electrical Schematic Drawing C10552216.

3.5 Performance. - With a potential of 16,000 volts dc, plus or minus 100 volts applied to the body assembly with image tube, the following requirements shall be met.

3.5.1 Image steadiness. - There shall be no visible flicker, no movement of the image nor audible sparking when observed for 1 minute. This observation shall be made after the expiration of 15 seconds tube stabilization time.

3.5.2 Collimation.

3.5.2.1 Elevation. - The line of sight through the reticle bore-sight mark shall form an angle of 71°30' plus or minus 4 minutes with a line through the centers of the two bores of the elevation input arm assembly when the line of sight is parallel to the horizontal mounting surface of the instrument within 1.0 mils, and the elevation input arm adjustment is set at midpoint. The midpoint of the elevation boresight adjustment is defined as that setting which allows a minimum remaining adjustment of 1/16 inch in each direction.

3.5.2.2 Deflection. - The line of sight through the reticle bore-= sight mark shall be parallel within 1 mil to a plane passing through the vertical geometric centerline of the sight body and the fixed center of the deflection boresight eccentric when the line of sight is parallel to the horizontal mounting surface within 10 mils. The instrument shall be accepted with the eccentric oriented and locked in a position such that a plane thru the vertical geometric centerlines of the sight body and the large diameter of the eccentric is parallel to this established line of sight within 1 mil.

3.5.3 Collimation change to to electron image shift. - After vibration, the periscope shall be subjected to ten cycles of "ON" and "OFF" rated voltage, the voltage being applied for 15 seconds and removed for 10 seconds. The collimation of the periscope at the expiration of

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Each "ON" period shall not exceed the requirements of 3.5.2.

3.5.4 Resolution. - Resolution as viewed on the paraxial axis of the viewing area tube shall be 3 minutes of arc or better.

3.5.5 Elevation tracking accuracy. - The angle between the line of sight through the reticle boresight mark and the line through the centers of the two bores of the elevation input arm (71°30') shall remain constant within plus or minus 1.0 mil, excluding backlash, at any setting of the arm between 15 degrees depression and 30 degrees elevation, and within plus or minus 1.5 mils at any setting above 30 degrees elevation up to and including 60 degrees. The elevation or depression angle is defined as the angle between the line of sight and the horizontal mounting surface of the instrument.

3.5.6 Backlash. - Backlash between the line of sight and the elevation input arm shall not exceed 0.20 mil at any setting of the arm.

3.5.7 Elevation excursion limits. - Actuation of the elevation input arm shall cause an angular change in the line of sight from 15 degrees depression to 60 degrees elevation, a total excursion of 75 degrees.

3.5.8 Alinement. - With the periscope mounting surface perpendicular to a plumb line, the following conditions shall be met.

3.5.8.1 Plumb travel. - After establishing coincidence with a plumb line at 60 degrees elevation, the line of sight shall not deviate from the plumb line by more than 1.0 mil, nor shall the total lateral spread exceed 1.0 mil, when the line of sight is depressed through the total excursion of 75 degrees.

3.5.8.2 Parallelism of reticle and image. - With the mounting surface of the periscope horizontal the 400 meter range line of the reticle shall be parallel to the image of a horizontal line within 30 minutes of arc.

3.5.8.3 Image tilt. - The image of a vertical target shall be vertical within one degree.

3.5.9 Focus.

3.5.9.1 Eyepiece focus. - The diopter scale shall indicate zero within 1/4 diopter when the image on the tube is brought into sharp focus. Diopter focus difference between image and reticle not to exceed 1/4 diopter.

3.5.9.2 Diopter movement. - The eyepiece shall be capable of movement throughout a range of plus or minus 3 diopters as read on the diopter scale.

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3.5.9.3 Backlash. - Backlash in the diopter movement shall not exceed 1/4 diopter.

3.5.9.4 Focusing knob. - Rotation of the focusing knob shall bring targets from 10 meters to infinity into focus.

3.5.10 Torque.

3.5.10.1 Input arm torque. - The torque required to rotate the input arm at standard ambient temperatures (60 to 90 degrees Fahrenheit) shall be as indicated on Drawing F10552282 and the torque shall not change from the value obtained at standard ambient temperatures by more than 5 pound inches when subjected to the temperatures of 3.3.2.

3.5.10.2 Prism rotation. - During expsoure at temperatures specified in paragraph 3.3.2 rotation of the input arm about its axis in either direction shall cause corresponding rotation of the head prism.

CAUTION: At no time shall the input arm be allowed to snap (spring return) against the travel stop.

3.5.10.3 Diopter adjustment torque. - The running torque required to rotate the diopter adjustment shall not exceed seven pound inches at standard ambient temperatures (60 to 90 degrees Fahrenheit) and the torque shall not change from the value obtained at standard ambient temperatures by more than three pound inches when subjected to the temperatures specified in 3.3.2.

3.5.10.4 Focusing knob torque. - The torque required to rotate the focusing knob shall not exceed 12 pound inches at standard ambient temperatures (60 to 90 degrees Fahrenheit).

3.5.11 Reticle illumination. - When illuminated by an insrument light, M50 type or equal, the reticle markings shall appear clearly defined when observed in a darkened room.

3.6 Sealing. - The periscope shall show no evidence of leakage in excess of 0.10 psig when subjected to an internal pressure of 5 psig for a period of five minutes. The interior atmosphere of the accepted sight shall contain nitrogen having a dewpoint at least as low as minus 25 degrees Fahrenheit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or Purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the

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specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article submission. - When a first article sample is required by the contract it shall be submitted to the Government for test and evaluation in accordance with the contract schedule. The sample shall consist of two periscopes. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, equipment, processes and including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply. After provisional acceptance at source, the initial production samples will be inspected and tested by the Government for all requirements and tests of the drawings and specifications. Government inspection and testing may be conducted at the contract facility or as otherwise specified by the contract. The Government resenes the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with Submission of Product provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and sub-assemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE I - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENTS	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65 Defective		
101.	Vibration	3.3.3	4.6.3
102.	Sealing	3.6	4.6.4
103.	Cleanliness	3.2.2	4.6.5
104.	Dielectric strength	3.4.1	4.6.6.1
105.	Insulation resistance	3.4.2	4.6.6.2
106.	Continuity	3.4.3	4.6.6.3
107.	Resolution	3.5.4	4.6.7
108.	Image steadiness	3.5.1	4.6.8.1
109.	Collimation, elevation	3.5.2.1	4.6.8.2
110.	Collimation, deflection	3.5.2.2	4.6.8.3
111.	Collimation change due to electron image shift	3.5.3	4.6.8.4
112.	Backlash	3.5.6	4.6.8.5
113.	Elevation tracking accuracy	3.5.5	4.6.8.6
114.	Elevation excursion limits	3.5.7	4.6.8.7
115.	Plumb travel	3.5.8.1	4.6.8.8
116.	Parallelism of reticle and image	3.5.8.2	4.6.8.9
117.	Image tilt	3.5.8.3	4.6.8.10
118.	Eyepiece focus	3.5.9.1	4.6.8.11
119.	Diopter movement	3.5.9.2	4.6.8.12
120.	Backlash, diopter movement	3.5.9.3	4.6.8.13
121.	Focusing knob	3.5.9.4	4.6.8.14
122.	Reticle illumination	3.5.11	4.6.10
123.	Torque, input arm	3.5.10.1	4.6.9.1
124.	Torque, diopter adjustment	3.5.10.3	4.6.9.2
125.	Torque, focusing Icnob	3.5.10.4	4.6.9.3
MINOR:	NONE DEFINED		

NOTE: The tests in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with “Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General. - One periscope shall be selected at random as a sample from each 100 produced. The sample shall meet the requirements and tests in Table II.

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TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawing visual
302.	General specification	3.2	MIL-F-13926 - visual

NOTE: The tests in Table 11 shall be conducted at a temperature between +60° and +90°F.

4.4.2 Environmental. - Three periscopes shall be selected at random as a sample from each 50 produced, or from each month's production, whichever occurs first. The sample shall meet the requirements and tests in Table III, and shall then meet the requirements and tests in Table I.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-65° and +155°F)	3.3.1	4.6.1
304.	Input arm torque	3.3.2, 3.5.10.1	4.6.2, 4.6.9.1
305.	Prism rotation	3.3.2, 3.5.10.2	4.6.2, 4.6.9.2
306.	Diopter adjustment torque	3.3.2, 3.5.10.3	4.6.2, 4.6.9.3
307.	Focusing knob torque	3.3.2, 3.5.10.4	4.6.2, 4.6.9.4

NOTE: The tests for Characteristics 304 through 307 inclusive shall be conducted at -40°F and at +l50°F.

4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended-by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by

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the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-confoming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- Special testing equipment to test sealing of the periscope assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.2 Performance of periscope (mounting). - Special testing equipment shall position the periscope assembly with its horizontal mounting surface in a horizontal plane within 20 seconds of arc. The necessary locating surfaces for positioning the persicope and locking features of a mounting collar shall be incorporated into the testing equipment to maintain a line of sight of the periscope, through the reticle boresight mark, as specified in 3.5.2.1. The testing equipment shall also be capable of securing the elevation input arm at an established zero elevation position (71°30'±4′), and calibrated angular positions required to measure the excursion range of the input arm from minus 15 degrees below the established zero position through 60 degrees above the established zero elevation position.

4.5.2.3.2.1 Collimation elevation tracking accuracy excursion limits, plumb travel, parallelism of reticle and image and image tilt of periscope assembly. - Target collimators utilized for these tests shall have a clear aperature of at least 2.0 inches and contain appropriate reticle patterns fir measuring the optical and mechanical accuracies of the periscope assembly. The collimator utilized for establishing the zero elevation position of the line of sight of the periscope and elevation input arm shall be aligned incident to the head prism of the periscope and auto-collimated to the mounting collar of the testing equipment utilized for positioning the periscope assembly. The calibrated target collimators to measure the periscope accuracies shall be alined at angular positions within a tolerance of ± 20 seconds. The alinement of the collimators shall include one at the minus 15 degrees, zero degree, plus 15 degrees, plus 30 degrees and plus 60 degrees angular positions. The vertical reticle lines

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of the target collimators shall be adjusted plumb, and alined coincident to a common plumb line image. The plumb line image shall be coincident, within 20 seconds, to a vertical plane that contains the vertical geometric centerlines of the periscope body assembly and the large diameter of the eccentric. Auto-collimation and alinement of the testing equipment shall be accomplished with the periscope assembly removed from the line of sight through the target collimators. The collimators shall be illuminated by filtered tungsten lamps. The filters shall not pass visible light rays below 0.65 microns.

4..6 Test methods and procedures.

4.6.1 Storage temperature. - The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.1. The periscope assembly shall be exposed to each temperature extreme specified in 3.3.1 for a minimum of 5 hours. After the temperature exposures, the periscope assembly shall be stabilized in an ambient temperature between +60° and 90° Fahrenheit and subjected to the appropriate visual, tactile, and audible examination of 4.6.3.1. The periscope assembly shall conform to the requirements of 3.3.1. Caution shall be exercised during this test to avoid subjecting the periscope to thermal shock.

4.6.2 Operating temperatures. - The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.2. The periscope assembly shall be exposed to each temperature extreme specified in 3.3.2 for a period of 5 hours. At the end of each 5 hour period, and while still at the temperature extreme, the periscope assembly shall be subjected to the applicable tests of 4.6.9.1 through 4.6.9.2 inclusive. The periscope assembly shall conform to the applicable requirements of 3.5.10.1 through 3.5.10.3 inclusive. Caution shall be exercised during this test to avoid subjecting the periscope to thermal shock.

4.6.3 Vibration. - Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.3.3 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in conformance with the Test Facilities" requirements of MIL-F-13926. Immobilize the elevation input arm of the periscope assembly to a position that shall prevent the arm from contacting the travel stops during the vibration cycle. The periscope assembly shall be vibrated in a vertical plane for a period of 5 minutes A 15 seconds. Upon completion of the vibration cycle the periscope assembly shall be subjected to the tests specified in 4.6.3.1.

4.6.3.1 Examination. - Subsequent to vibration, the perisocpe assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine conformance with 3.3.3, prior to being subjected to the remaining tests specified herein. A visual

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examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts such as diopter knob, focusing knob, and elevation input arm, etc. An audible examination shall be made of the periscope assembly by shaking it by hand and listening for loose internal parts or broken optical components, such as lenses, prism, reticle, etc. Any one of the possible defects defined above shall be cause to consider a periscope assembly defective. Periscope assemblies which have passed the above examinations shall be subjected to *the* tests specified in 4.6.4 to 4.6.10 inclusive to determine overall compliance with 3.3.3.

4.6.4 Sealing. - The internal pressurizing of the periscope assembly to test sealing shall be performed with special testing equipment conforming to 4.5.2.3.1. Pressurize the periscope assembly with dry nitrogen having a dewpoint at least as low as minus 25 degrees Fahrenheit in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the periscope assembly for a period of 5 minutes A 15 seconds. At the end of the 5 minute period there shall be no evidence of leakage, as indicated on the pressure gage of the testing equipment to determine compliance with 3.6. Upon satisfactory completion of the sealing test the periscope assembly shall be bled to 1 psig and flushed with dry nitrogen gas through a regulator that will prevent the internal pressure from exceeding the 1 psig during the flushing operation. The flushing shall continue until the nitrogen gas bled from the periscope has a dewpoint at least as low as minus 25 degrees Fahrenheit, as measured with a dewpoint tester. The periscope shall then be sealed. The internal atmosphere of the periscope shall conform to the requirements of 3.6.

4.6.5 Cleanliness. - Perform this test with diffuse illumination from a 0.65 micron tungsten filtered lamp presented at the entrance window of the periscope. Energize the periscope. Visually inspect the periscope through the eyepiece assembly. The periscope shall conform to the requirements of 3.2.2.

CAUTION: Prolonged exposure of the electron tube to direct rays of white light shall be avoided.

4.6.6 Electrical check. - Perform the following tests with the components specified in 3.4 removed from the circuitry. This is an in-process test and the inspection record for each periscope shall indicate the results for each test.

4.6.6.1 Dielectric strength. - The periscope shall be tested in accordance with Method 301 of MIL-STD-202, utilizing standard measuring equipment to determine conformance with the requirements of 3.4.1.

4.6.6.2 Insulation resistance. - Upon satisfactory completion of the dielectric strength test in 4.6.6.1, the periscope shall be subjected to the insulation resistance test in accordance with Method 302 of MIL-STD-202. Measurements between insulating points and ground shall be made utilizing standard measuring equipment to determine conformance with the

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requirements of 3.4.2.

4.6.6.3 Continuity. - A continuity test of the periscope shall be made utilizing standard measuring equipment to determine conformance to the electrical wiring and schematic drawings specified in 3.4.3.

4.6.7 Resolution. - The test for resolution shall be performed utilizing a resolving power chart on a wall target representing the angulalar subtense for minutes of arc specified in 3.5.4. Construction of the resolving power chart shall be as specified in the "Resolution Test" of MIL-O-13830 and contain four line sets as shown on Figure 1 therein. The target shall be illuminated by means of a filtered tungsten lamp. The filter shall not pass visible light rays below 0.65 microns. Energize the periscope assembly from a power source capable of supplying and maintaining the voltage specified in 3.5. The image tube shall be permitted a stabilization period of 15 seconds prior to conducting the resolution test of the periscope assembly. Upon completion of the stabilization period, view the target image through the periscope eye-piece assembly with the unaided eye. Obtain best focus of the numeral in the center of the resolving power chart by means of the periscope assembly diopter adjustment. At this point of focus, all four meridaians of the resolving power chart shall have the detectable line structure and line count to conform to the requirements of 3.5.4.

4.6.8 Performance of periscope. - The following tests shall be performed with the persicope assembly positioned on special testing equipment conforming to 4.5.2.3.2. Apply the operating voltage as specified in 4.6.7 to the periscope. The image tube shall be permitted a stabilization period of 15 seconds prior to conducting optical tests of the periscope assembly. Illuminate the periscope reticle pattern (see 4.6.10).

4.6.8.1 Image steadiness. - Position an auxiliary telescope at the eyepiece of the periscope assembly. The auxiliary telescope shall contain a crossline reticle and be capable of focusing on the screen of the image tube of the periscope. Observe through the auxiliary telescope and obtain coincidence of the horizontal and vertical reticle lines with respect to the horizontal and vertical reticle lines of the target image on the screen of the image tube. Observe the target image for a time period of not less than one minute. During this one minute time period there shall be no visible flicker or movement of the target image as specified in 3.5.1. Image movement shall be determined by means of the target image with respect to the auxiliary telescope reticle pattern.

4.6.8.2 Collimation, elevation. - Measure the midpoint setting of the elevation input arm adjustment with a scale. The midpoint adjustment of the elevation input arm shall meet the requirements of 3.5.2.1. Secure the elevation input arm in the zero elevation position (71°30') by means of the test equipment. Observe the target image through the periscope eyepiece assembly. The line of sight through the reticle

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boresight mark shall meet the collimation requirements specified in 3.6.2.1, as measured with reference to the target image.

4.6.8.3 Collimation - deflection. - Conduct this test in sequence with the test in 4.6.8.2. The line of sight through the reticle boresight mark shall meet the requirements in 3.5.2.2 for deflection, as measured with reference to the target image.

4.6.8.4 Collimation change due to electron image shift. - Conduct this test in conjunction with the tests in 4.6.8,2 and 4.6.8.3. The time cycling period and rated voltage shall be applied to the periscope as specified in 3.5.3. Collimation of the periscope at the end of each "ON" period shall meet the requirements of 3.5.3.

4.6.8.5 Backlash. - Observe through the periscope eyepiece assembly. With backlash excluded for the direction of travel of the elevation input arm, elevate the line of sight of the periscope to the established zero elevation position. Record the position of the reticle boresight mark as measured on the target image. Rotate the input arm of the periscope approximately 10 degrees in the same direction of travel, then with backlash included, reverse the rotation of the input arm and re-establish the zero elevation position without overtravel, Record the position of the reticle boresight mark as measured on the target image. Any difference between the recorded readings is the amount of backlash. Backlash shall then be measured at plus 30 degrees and 60 degrees in elevation and from the zero elevation position to minus 15 degrees in depression utilizing the test procedure above. Backlash shall meet the requirements specified in 3.6.7.

NOTE: During this test (4.6.8.5) if overtravel occurs, the test shall be nullified and the procedure specified for backlash shall again be performed.

4.6.8.6 Elevation tracking accuracy. - Conduct this test in sequence with the test in 4.6.8.5. Elevate the line of sight of the periscope, with backlash excluded to the zero elevation position as indicated on the test equipment. The reticle boresight mark shall be superimposed with respect to the established collimation position in 4.6.8.2. Depress the line of sight beyond the 15 degree depression position to eliminate backlash for elevation excursion. Rotate the elevation input arm to elevate the line of sight. At the minus 15 degrees depression and at any setting of the elevation input arm between minus 15 degrees depression and plus 30 degrees elevation, the line of sight through the boresight mark shall meet the requirements specified in 3.5.5. At any setting above 30 degrees elevation, up to and including 60 degrees elevation, the line of sight through the boresight mark shall meet the requirements specified in 3.5.5.

4.6.8.7 Elevation excursion limits. - Conduct this test in conjunction with the test in 4.6.8.6. Depress the line of sight of the

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periscope to minus 15 degrees depression. The line of sight through the boresight mark shall be in coincidence with respect to the target image as established in 4.6.8.6. Rotate the elevation input arm to elevate the line of sight 60 degrees above the established zero position. Angular displacement of the line of sight, due to actuation of the elevation input arm, shall meet the requirements specified in 3.5.7, as measured with respect to the target image.

4.6.8.8 Plumb travel. - Perform this test with the line of sight of the periscope elevated 60 degrees above the established zero position. The center of the boresight cross of the periscope reticle shall be in coincidence with respect to the vertical target image line. Rotate the elevation input arm to depress the line of sight to each known target image. The center of the boresight cross shall track the vertical target image lines within the tolerance and total excursion specified in 3.5.10.1, as measured with respect to the target image.

4.6.8.9 Parallelism of reticle and image. - With the line of sight horizontal, position one extremity of the indicated reticle line coincident to the horizontal target image line. The reticle line shall be parallel to the target image within the tolerance specified in 3.5.8.2, as measured at the other extremity of the horizontal reticle line with respect to the target image.

4.6.8.10 Image tilt. - Perform this test with the aid of an auxiliary telescope containing a vertical and horizontal reticle line. Plumb the vertical line of the telescope and position the telescope on a qualified leveled surface at the periscope eyepiece assembly. Observe through the telescope and superimpose the telescope vertical reticle line on the vertical target image line. The image of the vertical target line shall be parallel to the telescope vertical reticle line within the tolerance specified in 3.5.8.3 as measured with respect to the target image.

4.6.8.11 Eyepiece focus. - A calibrated dioptometer shall be utilized for measuring the accuracy of zero diopter setting of the diopter adjustment. Set the dioptometer diopter scale to zero, and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set the diopter adjustment of the periscope assembly to indicate zero diopter. Position the dioptometer at the periscope eyepiece. While viewing through the dioptometer, obtain best focus of the phosphor grain on the screen of the image tube by rotation of the periscope diopter adjustment. The phosphor grain on the screen of the image tube shall be in best focus within the tolerance specified in 3.5.9.1, as measured on the scale of the diopter adjustment. Record the periscope diopter adjustment setting. While viewing through dioptometer, obtain best focus of the periscope reticle on the screen of the image tube by rotation of the periscope dipter adjustment. Record the periscope diopter adjustment setting. The difference between the diopter adjustment settings for the image screen and the reticle shall not exceed the tolerance specified in 3.5.9.1.

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4.6.8.12 Diopter movement. - Position the diopter adjustment to zero diopter as indicated on the diopter scale with reference to the index mark. Rotate the diopter adjustment to its clockwise stop, note diopter setting on the diopter scale, then rotate the diopter adjustment to its counterclockwise stop, note diopter setting on diopter scale. The diopter movement shall conform to the requirements of 3.5.9.2.

4.6.8.13 Backlash - diopter movement. - With backlash excluded for direction of travel, rotate the diopter adjustment clockwise and obtain best focus of the phosphor grain on the screen of the image tube. Record the dipter scale reading. Continue rotation of diopter adjustment to its stop, then with backlash included, rotate the adjustment counter-clockwise and obtain best focus of the phosphor grain on the screen of the image tube without overtravel. Record diopter scale reading. The difference between the first and second recorded scale readings is the amount of backlash in the diopter movement. Backlash in the diopter movement shall not exceed the tolerance specified in 3.5.9.3. Should overtravel occur during rotation of the knob the test is nullified. When overtravel occurs the test for backlash shall be repeated.

4.6.8.14 Focusing range. - The test for short range focus requirements of the periscope shall be performed with the target image adjusted to a focus of 10 meters. With the diopter adjustment positioned for best focus of the phosphor grain on the screen of the image tube, rotate the focus knob of the periscope to short range setting. The periscope shall conform to the requirements specified in 3.5.9.4. The test for infinity (long range) requirements of the periscope shall be performed with the target image adjusted for infinity focus. Rotate the focusing knob to the infinity setting, the periscope shall conform to the requirements specified in 3.5.9.4..

4.6.9 Torque. - The tests in 4.6.9.1 through 4.6.9.4 shall be performed with standard measuring equipment equipped with appropriate adapters to accommodate the assembly under test.

CAUTION: At no time shall the input arm be permitted to strike the travel stops.

4.6.9.1 Input arm. - With the adapter affixed to the elevation input arm, the input arm shall be rotated through the range specified in 3.5.10.1. The running torque shall not be less than or exceed the values specified in 3.5.10.1 at the temperature indicated.

4.6.9.2 Prism rotation. - Conduct this test in sequence with the test in 4.6.9.1. Rotate the elevation input arm about its axis in either direction. A visual inspection of the head prism through the head window shall be performed while the periscope assembly is exposed at the temperatures indicated in 3.3.2. During this rotation there shall be no binding or restriction of travel of the head prism as specified in 3.5.10.2.

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4.6.9.3 Diopter adjustment. - With the adapter affixed to the diopter movement, the diopter movement shall be rotated through the range specified in 3.5.10.3. The running torque shall not be less than or exceed the values specified in 3.5.10.3 at the temperatures indicated.

4.6.9.4 Focusing knob. - Conduct this test in sequence with the test in 4.6.9.3. The running torque required to rotate the focusing knob through the excursion specified in 3.5.10.4 shall not be less than or exceed the values specified in 3.5.10.4 at the temperatures indicated.

4.6.10 Reticle illumination. - A device consisting of a .19 ampere lamp energized with 3.0 volts dc shall be utilized to illuminate the periscope reticle. An appropriate mask shall be utilized with the device to aid in the projection of the illumination through the Filter 10552170 of the periscope. Illuminate the reticle of the periscope with the device. The illuminated reticle etchings shall be clearly defined when visually examined against a darkened background. Should the brightness of a reticle defect be greater than the reticle lines, the defect shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 10514542. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The periscope is used *as* a night sighting device for a machine gun in the cupola of the armored Assault Vehicle XM729.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.

Custodian:

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